

GALLIVARE LAPLAND, Sweden

WMO: 020490

Lat: 67.1407N

Lon: 20.6413E

Elev: 1178

StdP: 14.08

Time Zone: 1.00 (EUC)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-24.0	-18.4	-30.4	1.0	-24.1	-24.0	1.5	-18.4	20.2	29.1	17.0	26.3	2.7	310	0.712

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.5	75.5	59.6	71.9	58.2	68.4	56.8	63.0	71.6	60.6	68.1	58.5	65.4	6.4	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
59.5	79.3	65.8	57.4	73.4	63.5	55.4	68.3	61.2	29.1	71.5	27.4	68.1	25.9	65.1	71.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 17.6	14.7	12.5	-31.1	80.4	5.9	4.4	-35.4	83.6	-38.9	86.1	-42.2	88.6	-46.6	91.8		
(5)			-31.4	65.6	5.9	2.5	-35.6	67.4	-39.1	68.9	-42.4	70.3	-46.7	72.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	31.6	8.7	9.8	18.5	29.9	41.4	51.3	57.4	53.1	44.2	31.7	18.5	12.9
	DBStd	19.45	13.38	13.82	10.62	8.17	7.17	6.55	5.56	5.81	6.24	8.08	10.85	12.58
	HDD50	7226	1280	1127	976	602	284	61	34	193	569	945	1149	
	HDD65	12214	1745	1547	1441	1052	733	413	244	371	625	1034	1395	1614
	CDD50	497	0	0	0	0	16	99	235	129	18	0	0	0
	CDD65	12	0	0	0	0	0	2	8	2	0	0	0	0
	CDH74	172	0	0	0	0	7	39	105	21	0	0	0	0
	CDH80	17	0	0	0	0	0	5	12	0	0	0	0	0
	WSAvg	5.3	4.3	4.8	6.3	6.0	6.3	6.2	5.3	4.9	5.5	4.9	4.1	4.4
	PrecAvg	23.3	1.5	1.3	1.1	1.1	1.7	2.5	3.7	3.0	2.2	1.8	1.9	1.7
(15) Precipitation	PrecMax	28.9	2.8	2.8	2.7	2.1	3.0	4.4	5.9	5.4	5.3	3.9	4.2	5.2
	PrecMin	16.1	0.1	0.3	0.2	0.1	0.1	0.3	1.6	1.1	0.6	0.2	0.3	0.3
	PrecStd	3.7	0.6	0.7	0.6	0.6	0.8	1.1	1.1	1.2	1.2	0.9	0.9	1.0
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	39.9	41.2	44.8	57.2	73.8	79.1	81.6	76.8	65.8	52.0	42.8	40.7	
	0.4% MCWB	35.8	36.7	37.6	45.9	54.2	59.3	64.5	62.5	55.7	46.9	40.1	36.6	
	2% DB	35.4	36.8	40.3	50.0	66.5	73.1	76.9	72.9	61.1	48.5	38.9	36.0	
	2% MCWB	32.6	33.5	35.0	40.2	52.0	55.8	61.4	59.9	52.9	45.1	37.2	33.5	
	5% DB	30.9	32.4	37.3	46.5	60.2	68.9	73.3	69.3	57.8	46.1	35.3	33.0	
	5% MCWB	29.4	30.0	33.2	38.5	48.8	55.0	59.4	58.9	51.5	43.0	33.8	31.6	
	10% DB	27.0	29.5	34.1	43.0	54.8	64.5	69.6	64.7	55.2	42.9	32.6	30.4	
	10% MCWB	26.0	28.0	30.6	36.6	44.9	52.7	58.1	56.3	50.0	40.2	31.8	29.3	
	0.4% WB	36.5	37.2	38.0	45.6	56.5	63.0	66.5	65.2	57.4	48.4	40.4	36.8	
	0.4% MCDB	39.4	40.7	43.5	56.1	70.1	74.1	76.4	72.8	62.5	50.2	42.0	39.5	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	32.7	33.3	35.4	41.6	52.8	59.0	63.9	62.2	54.9	45.7	36.9	33.9	
	2% MCDB	34.8	36.1	39.5	48.5	64.5	67.2	73.1	69.7	59.2	48.1	38.7	35.7	
	5% WB	29.4	30.3	33.3	39.1	49.2	56.3	61.7	59.7	52.9	43.0	33.8	31.7	
	5% MCDB	30.8	32.0	36.9	44.8	58.7	65.0	69.6	66.6	56.7	45.5	34.8	32.8	
	10% WB	26.0	27.8	30.8	37.0	46.2	54.1	59.6	57.4	51.0	40.5	32.0	29.2	
	10% MCDB	27.0	29.4	33.8	42.4	53.2	62.9	66.7	63.4	54.3	42.5	32.6	30.3	
(35) Mean Daily Temperature Range	MDBR	16.5	18.2	20.5	18.6	17.0	16.9	17.5	17.6	15.9	12.1	13.5	15.6	
	5% DB MCDBR	17.4	16.1	18.8	21.5	27.5	24.8	24.7	24.9	20.5	15.2	12.4	14.6	
	5% DB MCWBR	16.1	14.0	14.7	13.6	14.9	11.6	11.8	13.8	13.4	11.7	11.1	13.4	
	5% WB MCDBR	16.2	15.0	17.9	19.9	24.7	20.5	20.5	21.8	17.3	13.7	11.4	14.2	
	5% WB MCWBR	15.1	13.2	14.4	13.0	13.9	11.0	11.0	12.9	12.0	11.0	10.5	13.4	
(40) Clear-Sky Solar Irradiance	taub	0.181	0.244	0.281	0.311	0.329	0.333	0.341	0.326	0.295	0.268	0.344	N/A	
	taud	1.932	2.132	2.213	2.258	2.451	2.498	2.487	2.511	2.574	2.377	2.793	N/A	
	Ebn at Noon	58	188	245	268	276	278	271	264	248	185	71	0	
	Edh at Noon	4	16	25	31	29	29	28	25	18	12	5	0	
(44) All-Sky Solar Radiation	RadAvg	18	179	647	1241	1558	1611	1507	1101	611	212	39	0	
	RadStd	1	19	39	57	143	125	148	95	82	34	4	0	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (42 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page